HW. Statistical methods for risk management. Kantemirova Elizaveta

№ 1

Let Y_i be the default indicator of firm i:

$$Y_i = \begin{cases} 1, \text{ if bond } i \text{ is default} \\ 0, \text{ otherwise} \end{cases} \qquad Y_i \sim \text{Bernoulli}(\underbrace{0.03}_{\text{probability of default}}) \text{ i.i.d.}$$ (it's Bernoulli, as only 2 outcomes (0 or 1) and it's random)

for Bernoulli: $\mathbb{E}[Y_i] = p = 0.03$, $\mathbb{V}ar[Y_i] = p \cdot (1-p) = 0.03 \cdot 0.97 = 0.0291$

Today $V_0 = 98$, after 1 year, $V_{1,i} = 104(1-Y_i)$ (104, if there is no default, and 0, if it appears, as recovery rate is 0), then:

$$L_i = V_0 - V_{1,i} = 98 - 104(1-Y_i) = 104Y_i - 6 \text{ or } L_i = \begin{cases} 98, \text{ default (probability} = 1-p = 0.97) \\ -6, \text{ otherwise (probability} = 0.03) \end{cases}$$

Loss of the portfolio: $L = \sum_{i=1}^{n} L_i = \sum_{i=1}^{N} (104Y_i - 6) = \boxed{104 \sum_{i=1}^{N} Y_i - 6N}$

Let's introduce M, as a quantity of defaults, then $M = \sum_{i=1}^{N} Y_i$, as Y_i are independent and $Y_i \sim$ Bernoulli (0.03), then $M \sim Bin(N, 0.03) \Rightarrow L = 104M - 6N$

then, $VaR_{0.95} = 104 q_{0.95}(M) - 6N$, by definition, $q_{0.95}(M) = \min\{k : P(M \le k) \ge 0.95\}$

$$P(M \le k) = P(M=0) + P(M=1) + \dots + P(M=k) = \sum_{j=0}^{k} P(M=j) \Rightarrow P(M=j) = \binom{N}{j}(0.03)^j (0.97)^{N-j} \Rightarrow P(M \le k) = \sum_{j=0}^{k} P(M=j) = \sum_{j=0}^{k} \binom{N}{j}(0.03)^j \cdot (0.97)^{N-j}$$

so, $q_{0.95}(M) = \min\{k : \sum_{j=0}^{k} \binom{N}{j}(0.03)^j \cdot (0.97)^{N-j} \ge 0.95\}$, for huge N, we can use $\underset{\text{Central Limit Theorem}}{CLT}$, $M \approx \mathcal{N}(Np, Np(1-p)) \Rightarrow M \approx \mathcal{N}(0.03N, 0.0291N) \Rightarrow L = 104M - 6N$ // also approx. norm. dist.

$$\mathbb{E}[L] = \mathbb{E}[104M - 6N] \overset{\text{by properties}}{=} 104\,\mathbb{E}[\underset{p \cdot N}{M}] - 6N = 104 \cdot 0.03 \cdot N - 6N = -2.88N$$ (as avg. loss is negative, avg. profit $\approx 2.88N$)

$$\mathbb{V}ar[L] = \mathbb{V}ar[104M - 6N] \overset{\text{by properties}}{=} 104^2 \cdot \mathbb{V}ar[\underset{N \cdot p(1-p)}{M}] = 104^2 \cdot N \cdot 0.03 \cdot 0.97 = 314.7456N$$

$$\sigma_L = \sqrt{\mathbb{V}ar(L)} = 104 \cdot \sqrt{0.03 \cdot 0.97N} \approx 17.741\sqrt{N}, \quad L \approx \mathcal{N}(-2.88N, 314.7456N)$$

for normal dist. we'll use $VaR_\alpha(L) = \mu_L + \sigma_L \cdot \Phi^{-1}(\alpha) \Rightarrow VaR_{0.95}(L) \approx -2.88N + 17.741\sqrt{N} \cdot 1.645 \approx \boxed{-2.88N + 29.18\sqrt{N}}$ (negative $-2.88N$ means that with prob-ity 95% portfolio'll make a profit (expected $2.88N$))

№2.

in this task we need to show that $VaR(X+Y) \leq VaR(X) + VaR(Y)$

Let $Z \sim N(0,1)$, then $X = aZ \Rightarrow q_\alpha(X) = \alpha\Phi^{-1}(\alpha)$, $z_\alpha = \Phi^{-1}(\alpha)$, then

by definition, for normal distribution, $VaR_\alpha(L) = \mu_\alpha + \sigma_\alpha \Phi^{-1}(\alpha)$

$$VaR_\alpha(X) = 0 + \overset{\sigma_X = \sqrt{Var(X)}}{a} \cdot z_\alpha = a \cdot z_\alpha$$

$$VaR_\alpha(Y) = 0 + \overset{\sigma_Y = \sqrt{Var(Y)}}{b} \cdot z_\alpha = b \cdot z_\alpha$$

$$\longrightarrow \quad VaR_\alpha(X) + VaR_\alpha(Y) = (a+b) \cdot z_\alpha$$

a sum of independent random variables is normal, then

$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y] = 0 + 0 = 0$

by definition, $Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$

from independence we get $Cov(X,Y) = 0 \Rightarrow Var(X+Y) = Var(X) + Var(Y) = a^2 + b^2 \Rightarrow X+Y \sim (0, a^2+b^2)$, then $VaR(X+Y) = \overset{\mu_{X+Y}}{0} + z_\alpha \cdot \overset{\sigma_{X+Y} = \sqrt{Var(X+Y)}}{\sqrt{a^2+b^2}} = z_\alpha \cdot \sqrt{a^2+b^2}$

for $a, b \geq 0$:

$$(a+b)^2 = a^2 + b^2 + 2ab \geq a^2 + b^2 \Rightarrow a + b \geq \sqrt{a^2+b^2} \quad \Big| \cdot \underset{\text{(it's for } \alpha > 0.5)}{z_\alpha = \Phi^{-1}(\alpha) > 0}$$

$$\overset{VaR_\alpha(X) + Var_\alpha(Y)}{z_\alpha(a+b)} \geq \overset{Var_\alpha(X+Y)}{z_\alpha \cdot \sqrt{a^2+b^2}}$$

$$\Downarrow$$

$$\boxed{VaR_\alpha(X+Y) \leq VaR_\alpha(X) + VaR_\alpha(Y)}$$

№3

by conditions, $\mu = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$ $\Rightarrow$ $\mathbb{V}ar[X_1] = \mathbb{V}ar[X_2] = 1$, $\mathbb{E}[X_1] = \mathbb{E}[X_2] = 0$, $Cov(X_1, X_2) = 0.5$, $V_0 = 100$

1)

for A):

$\omega_{A_1} = \omega_{A_2} = \frac{1}{2}$

from lecture. $L^{\Delta}_{t+1} = -V_t \cdot \sum_{i=1}^{d} \omega_{t,i} X_{t+1,i}$ $\Rightarrow$ $L^{\Delta}_A = -100 \cdot \left(\frac{1}{2}X_1 + \frac{1}{2}X_2\right) = \boxed{-50(X_1 + X_2) \text{ or } L^{\Delta}_A = -50X_1 - 50X_2}$

for B):

$\omega_{B_1} = 1$, $\omega_{B_2} = 0$

from lecture. $L^{\Delta}_{t+1} = -V_t \cdot \sum_{i=1}^{d} \omega_{t,i} X_{t+1,i}$ $\Rightarrow$ $L^{\Delta}_A = -100 \cdot (1 \cdot X_1 + 0 \cdot X_2) = \boxed{-100X_1}$

2)

$\mathbb{V}ar(X_1 + X_2) = \mathbb{V}ar(X_1) + \mathbb{V}ar(X_2) + 2Cov(X_1, X_2) = 1 + 1 + 2 \cdot 0.5 = 3$ $\Rightarrow$ $X_1 + X_2 \sim N(0, 3)$

then, $L^{\Delta}_A = -50(X_1 + X_2) \sim N(0, 50^2 \cdot 3)$ $\Rightarrow$ $L^{\Delta}_A \sim N(0, 7500)$ (as $\sigma_A^2 = 7500 \Rightarrow \sigma_A = 50\sqrt{3}$)

by definition, for normal distribution, $VaR_A(L) = \mu_A + \sigma_A \Phi^{-1}(L) = 0 + 50\sqrt{3} \cdot 2.33 = 50\sqrt{3} \cdot 2.33$

$L^{\Delta}_B = -100X_1$, as $X_1 \sim N(0,1)$, then $L^{\Delta}_B \sim N(0, 100^2)$ (as $\sigma_B^2 = 100^2$, then $\sigma_B = 100$)

by definition, for normal distribution, $VaR_B(L) = \mu_B + \sigma_B \Phi^{-1}(L) = 0 + 100 \cdot 2.33 = 100 \cdot 2.33$

$\frac{VaR_{0.99}(L^{\Delta}_A)}{VaR_{0.99}(L^{\Delta}_B)} = \frac{50\sqrt{3} \cdot 2.33}{100 \cdot 2.33} = \boxed{\frac{\sqrt{3}}{2} \approx 0.866}$

3)

Let $X_1^{(1)}$ - log return of 1st stock during 1st day, $X_2^{(2)}$ - log return of 2nd stock during 2nd day; by conditions of task returns of different days are independent $\Rightarrow$

$X_1^{(1)}$ and $X_2^{(2)}$ are independent. Also, $X_1^{(1)} \sim N(0,1)$, $X_2^{(2)} \sim N(0,1)$

In the beginning of 1st day all capital (100) was invested in 1st stock, for log return $S_{t+1} = S_t \cdot e^X$, so at the end of 1st day $V_1 = 100e^{X_1^{(1)}}$

During 2nd day, $V_2 = V_1 \cdot e^{X_2^{(2)}} = 100 \cdot e^{X_1^{(1)}} \cdot e^{X_2^{(2)}} = \boxed{100 \cdot e^{X_1^{(1)} + X_2^{(2)}}}$, let $Z = X_1^{(1)} + X_2^{(2)}$ $\Rightarrow$ $V_2 = 100e^Z$

As return belong to different days, then they're independent $\Rightarrow$ $Cov(X_1^{(1)}, X_2^{(2)}) = 0$ $\Rightarrow$ $Var(Z) = 1 + 1 + 0 = 2$ $\Rightarrow$ $Z \sim N(0,2)$

by definition, $L = -(V_2 - V_0) = 100 - V_2 = 100 - 100e^{Z} = \boxed{100 - 100e^{Z}}$

Let $q = VaR_{0.99}(L)$, as dist. is continuous, $P(L \leq q) = 0.99$

$$P(100(1-e^{Z}) \leq q) = 0.99$$

$$P\left(e^{Z} \leq 1 - \frac{q}{100}\right) = 0.99$$

$$P\left(Z \geq \ln\left(1 - \frac{q}{100}\right)\right) = 0.99$$ (as $Z \sim N(0,2)$, $\sigma_Z = \sqrt{2} \Rightarrow q_{0.01}(Z) = \sqrt{2}\,\Phi^{-1}(0.01)$. By symmetry, $\Phi^{-1}(0.01) = -\Phi^{-1}(0.99) \approx -2.33$)

then, $\ln\left(1 - \frac{q}{100}\right) = -2.33\sqrt{2} \Rightarrow \frac{q}{100} = 1 - e^{-2.33\cdot\sqrt{2}} \Rightarrow VaR_{0.99}(L) = \boxed{100\left(1 - e^{-2.33\sqrt{2}}\right) \approx 96.3}$

from lecture, $e^{Z} \approx 1 + Z \Rightarrow L^{\Delta} = 100\,(1 - (1+Z)) = -100Z \Rightarrow \boxed{L^{\Delta} = -100\left(X_1^{(1)} + X_2^{(1)}\right)}$

$-100Z \sim N(0, 100^2 \cdot 2) \Rightarrow L^{\Delta} \sim N(0, 20000) \Rightarrow \sigma_{L^{\Delta}} = 100\sqrt{2} \Rightarrow \boxed{VaR_{0.99}(L^{\Delta}) = 2.33 \cdot 100\sqrt{2} \approx 329.5}$

Well done!